

System Requirements Specification (SRS)

1. Introduction

This document specifies the system requirements for a Company Management System, based on the provided Entity-Relationship Diagram (ERD).

2. System Description

The system is designed to manage the core entities of a company, including employees, departments, projects, and dependents. It establishes clear relationships, constraints, and management rules as illustrated in the ERD.

3. Functional Requirements

- The system shall store and manage employee information, including a unique ID, composite name (First & Last name), address, gender, birthdate, and join date.
- The system shall allow assigning employees to specific departments (Works_For relationship) and record the start date of their employment in that department.
- The system shall track department management, identifying one manager for each department and recording the date they started their managerial role (Since).
- The system shall manage department details, including a unique name, unique department number (D_NO), and multiple office locations.
- The system shall track project details, such as project number (P_NO), name, and a single specific location.
- The system shall record hours worked by employees on projects (Works_on relationship), supporting a many-to-many relationship.
- The system shall maintain supervisory relationships, allowing an employee to be designated as a supervisor for other employees.
- The system shall manage dependent information for employees, including the dependent's name, gender, and relationship to the employee. (Note: Dependents are treated as weak entities).
- The system shall calculate derived attributes, specifically the total number of employees per department (No_employee).

4. Non-functional Requirements

- **Data Integrity:** The system shall ensure that all primary keys (ID, D_NO, P_NO) are unique and that weak entities (Dependents) are correctly linked to their owners.
 - **Performance:** The system shall provide efficient query performance when retrieving department hierarchies and project assignments.
 - **Scalability:** The system shall be capable of handling an increasing number of employees and historical project data.
 - **Consistency:** The system shall maintain referential integrity across all relationships, especially during the deletion of parent records.
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